# Multilingual NLP in the LLM Era

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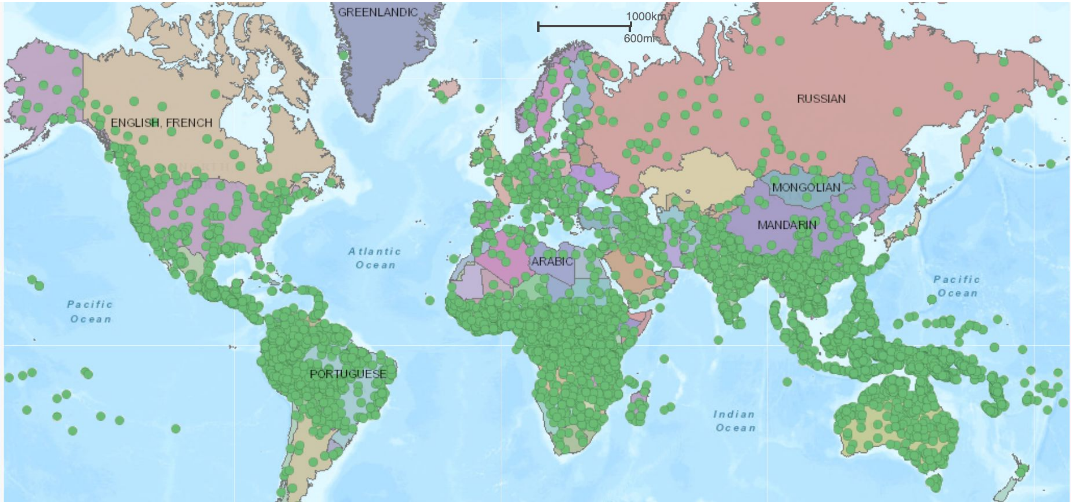
# Today's Outline

- Why multilingual NLP?
- Approaches to multilingual NLP: two paradigms
  - Route 1: per-language and adapted monolingual models
  - Route 2: jointly trained multilingual models
- The standard recipe for Multilingual LMs
  - Multilingual data
  - Tokenization
  - Joint pretraining
  - Cross-lingual transfer and adaptation
- Evaluation: what we measure and what we don't
- Open problems

# Why Multilingual NLP?

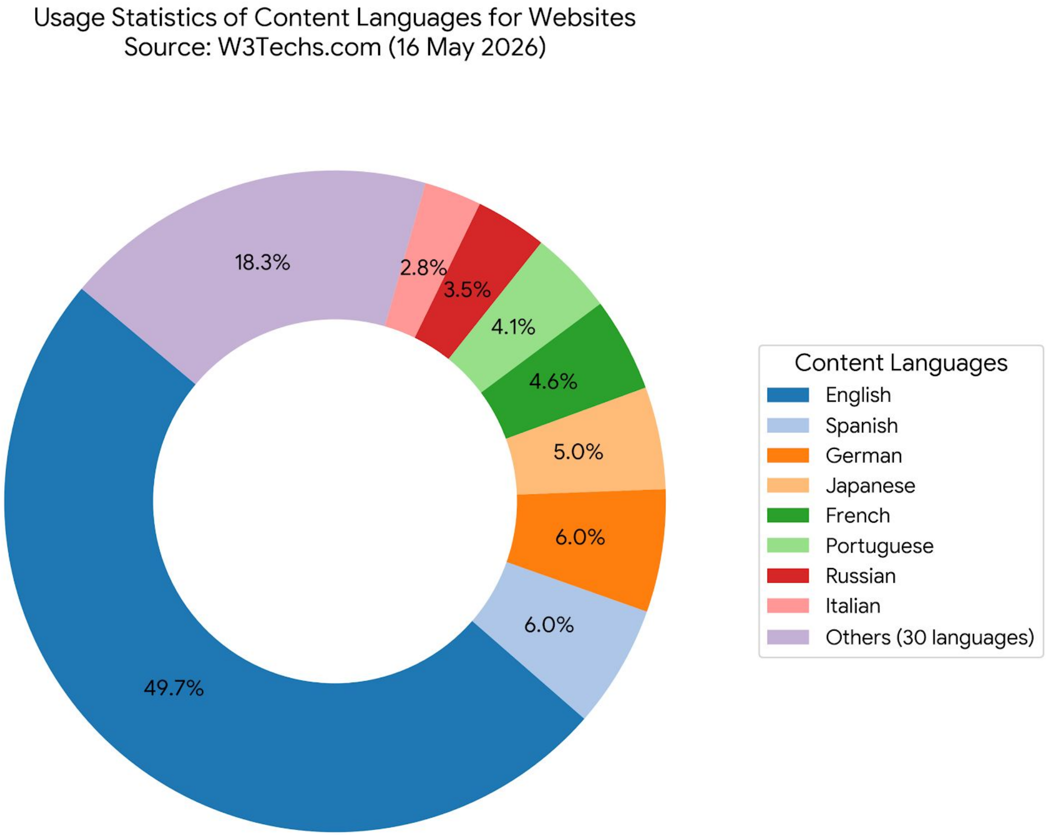



World's Languages

Language in the Real World

- ~7,170 living languages worldwide[1]
- ~150–200 with more than 1M speakers[2]
- 43% of people are bilingual, ~13% speak three or more languages[3]

# Why Multilingual NLP?

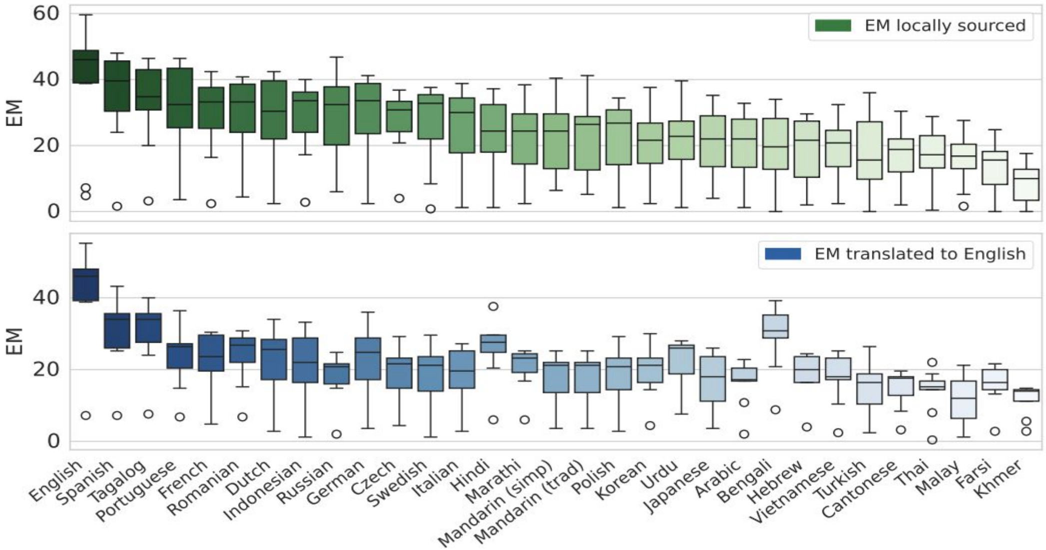



Language on the Internet

- ~49% of web content is in English, despite English being spoken (mostly as L2) by under 20% of the world[4]
- ~340 languages have a Wikipedia edition[5], with a large skew of the content belonging to the top 10

# Why Multilingual NLP?

Exact Match (EM) on MultiLoKo: Multilingual Local Knowledge Benchmark




*Frontier LLMs show 15–30% accuracy gaps between English and African languages on reasoning tasks[6], and 20+ point gaps between best- and worst-performing languages on local-knowledge benchmarks[7]*

## Language in NLP and ML

- ~88% of the world's ~7,000 languages do not have sufficient labeled or unlabeled resources for support in modern NLP pipelines
- Frontier multilingual LLMs now claim coverage of 100+ languages (Aya-101: 101 languages; BLOOM: 46 languages; NLLB-200: 200 languages for MT). Yet the performance gap between English and other languages remains large

# Why Multilingual NLP?

**Consequences + Implications**

- Language technology is increasingly the interface to information, education, healthcare, and economic opportunity. Every unsupported language *prevents some subset of people from accessing these resources*.
- At a more subjective level: models trained predominantly on Western, English-language data reflect cultural and regional biases that alienate users.
- In sum: "Supports N languages" on a model card ≠ what users in those languages actually experience.
- → **If we want tools that are genuinely useful for the world, multilingual support cannot be an afterthought**

**Rest of this lecture**: how the field tries to do this, where it works, where it breaks, and what's still open

# Approaches to Multilingual NLP

# Overview: Two paradigms for Multilingual NLP

| | Route 1: Monolingual NLP in many languages | Route 2: Cross-lingual NLP |
|---|---|---|
| Idea | Strong monolingual models: either (a) Train language-specific models from scratch or (b) Adapt a high-resource base model | One multilingual model: trained jointly on all languages from the start |
| What it exploits | | |
| Strengths | | |
| Limitations | | |
| Examples | | |

# Route 1: Monolingual NLP in many languages

## Motivation

Pretraining and labeled data are abundant for a handful of languages. Can build individual models for those, but building from scratch for 7,000+ is not feasible.

## Strategies

### Train from scratch per language

- **CamemBERT** French (FR)
- **FinBERT** Finnish (FI)

### Adapt a high-resource base model

**English base LLM**
*Llama, Mistral, …*

→

- **Sabiá** Portuguese (PT)
- **AceGPT** Arabic (AR)
- **Swallow** Japanese (JA)

# Route 1: Monolingual NLP in many languages

## Adapting a high-resource base model

Many language-specific multilingual models follow this route: take a strong (typically English-dominant) base and add target-language capability via continued pretraining. The practical question: how do we add languages to a base model without breaking what is there?

Several points in the pipeline where language adaptations can be done:

1. **Tokenizer adaptation** — Vocabulary extension →
2. **Continued pretraining** — Source + target mixed →
3. **Multilingual SFT** — Instruction-response data →
4. **Multilingual DPO** — Preference pairs

### Stage 2
**Continued pretraining**

- Naive continued pretraining degrades performance on base language; mitigating this forgetting is the point of tension.
- Solution: Parameter-efficient methods
  - Language adapters, e.g., MAD-X (still used at smaller scales)
  - LoRA: now the default. Layer-selective LoRA: trainable parameters placed only where language-specific processing happens

### Stage 3
**Instruction tuning across languages**

- Native instruction data exists for few languages
- Most multilingual SFT data is translated from English ⟶ issues like translationese, cultural assumptions

# Route 1: Monolingual NLP in many languages

## Limitations

- Substantial (labeled) target-language data still required
- Adaptation tends to cause (catastrophic) forgetting of the source language
- Cross-lingual tasks (e.g. machine translation) don't benefit from shared model; need separate setup
- Effort doesn't compound: adapting the model for the 30th language is as much work as the 3rd

# Route 2: Cross-lingual NLP

## The standard recipe for multilingual LMs

1. **Multilingual data**
   Acquiring, labeling, cleaning
2. **Tokenization**
   Shared vocabulary
3. **Joint pretraining**
   All languages together
4. **Adaptation**
   Zero-shot performance and target-language tuning

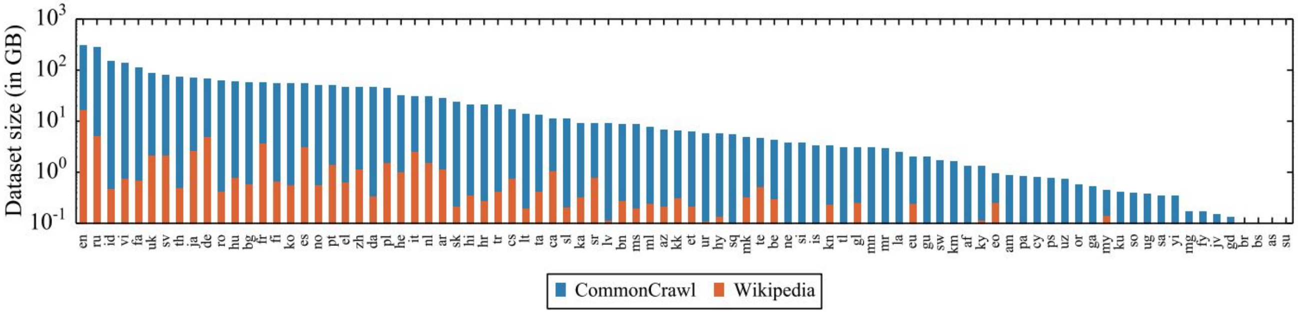
# Step 1: Multilingual Corpora

**Basic Idea:** Acquire a large multilingual corpus and train on it

**Where It Falls Apart:**

Available Resources

- Coverage has grown rapidly in recent years: CC-100 → mC4 → CulturaX → FineWeb-2 (~2000 languages). But coverage is far from equally distributed (stat from earlier: ~88% of languages have **essentially no web text!)**
- Systematic biases are in place
  - Most data reflects US-based crowdwork conventions
  - Religious texts often the only domain for low-resource languages




Amount of data in GiB (log-scale) for the 88 languages that appear in both Wikipedia and CommonCrawl

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Quality Matters as Much as Coverage

- ~20% of languages in major corpora are <50% acceptable[9] (both quality and LID issues)[10]
- Cleaning pipelines designed for English systematically harm other languages (e.g., special character handling); standard filters drop ~60% of articles in many low-resource Wikipedias[11]

Open Problems

- Corpora filtered with imperfect language identification (LID); errors are systematic, not random, and hurt low-resource languages the most

# Step 2: Tokenization

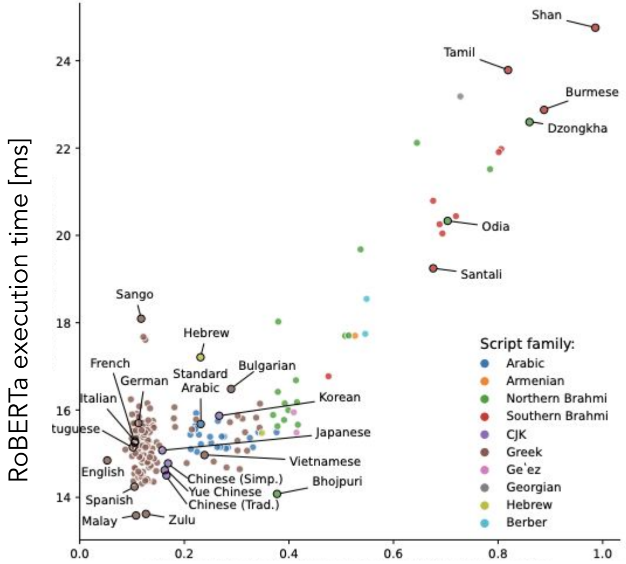
**Basic Idea:** Take (a sample of) your multilingual training data and train a new tokenizer on it

**Where It Falls Apart:**

- Tokenization algorithms designed/optimized for high-resource language → mishandling of many other languages
- Joint BPE / UnigramLM over imbalanced data allocates tokens disproportionally to

Both issues lead to: over-segmentation or nonsensical segmentation of low-resource and morphologically complex languages.

The same content in Tamil takes ~9× the tokens of English.[12]




Tokenization length for the FLORES-200 parallel corpus

Why this matters

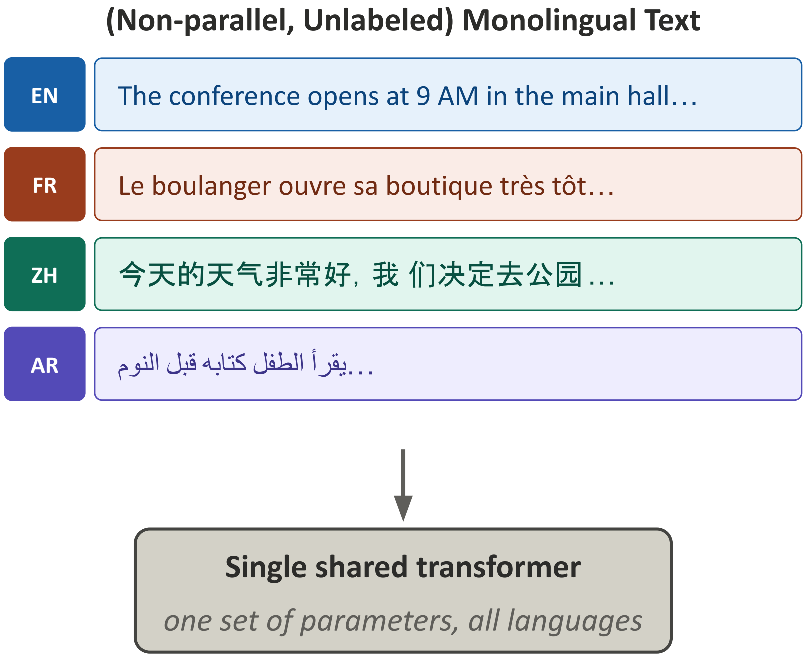
- Smaller effective context windows for non-English users
- Higher API costs and slower inference
- Users of low-resource languages pay more for the same information

Mitigation strategies

- Upsample low-resource languages during tokenizer training
- Byte-level fallbacks
- Cross-lingually aware tokenization, e.g. Parity-Aware BPE

# Step 3: Joint pretraining

**Basic Idea:** Standard joint pretraining on the combined multilingual corpus. No parallel data, no translation signal.




**Observations:**

- Yields a multilingual base model whose representations behave consistently across languages. Why does this work without explicit cross-lingual supervision?
    - Shared parameters force the model to encode concepts efficiently, so it exploits cross-lingual similarities
    - Shared vocabulary gives shared embeddings for digits, names, root words, and code-switched tokens

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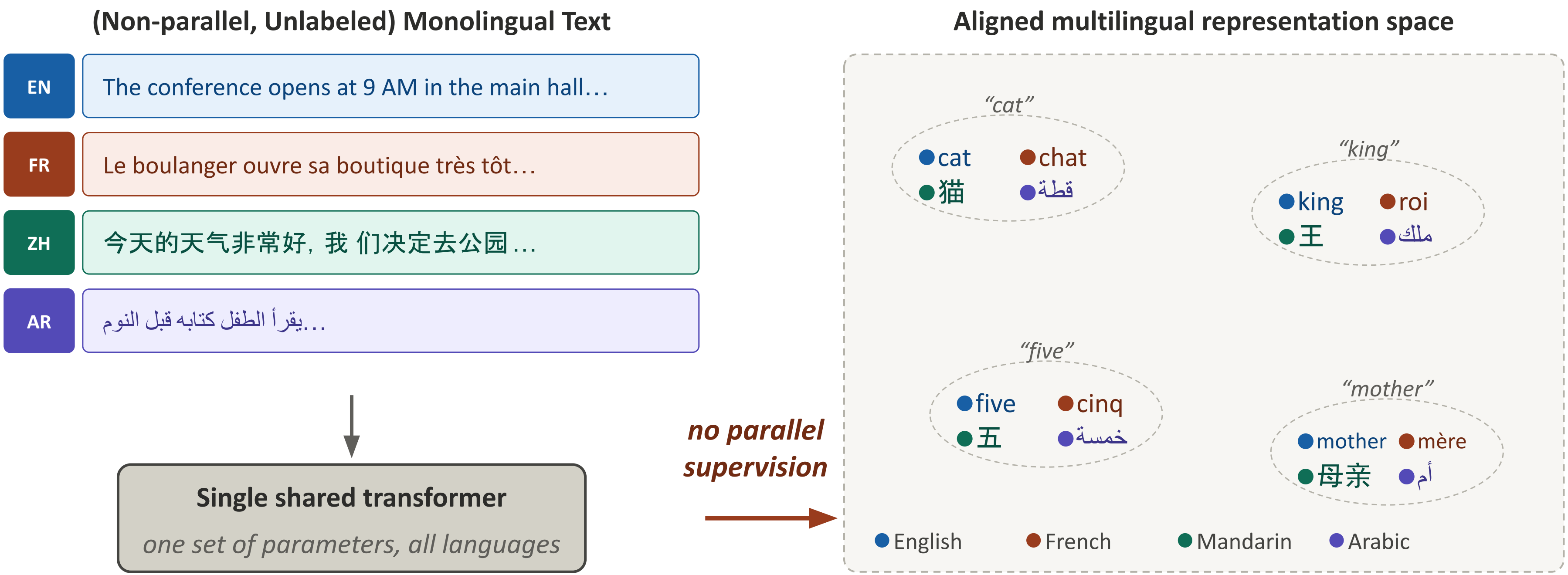

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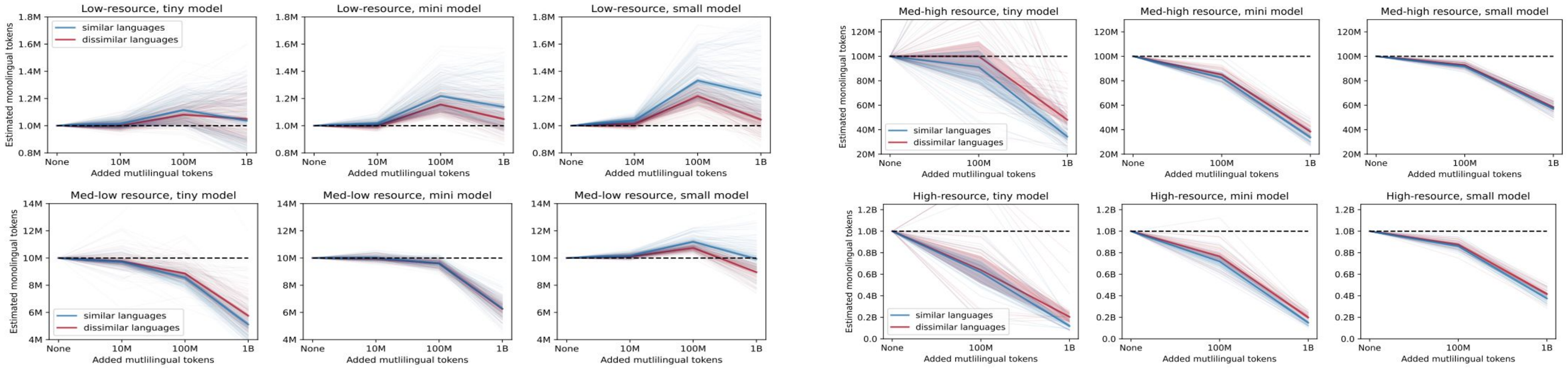
**Where It Falls Apart:**

- "The curse of multilinguality": Representation capacity of a model is limited. Adding more data from one language may take resources away from another, harming performance on it[13]



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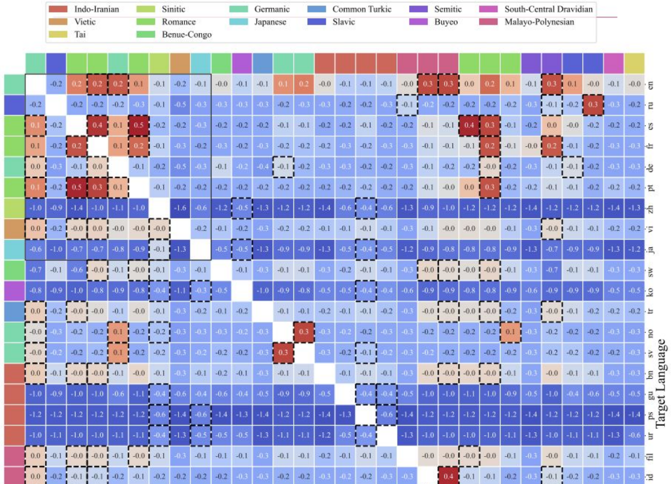
**Where It Falls Apart:**

- Choosing data mixtures becomes $N^2$ times more complicated. Languages hurt and help each other (cross-lingual transfer/interference) and all these interactions should be accounted for.[14]



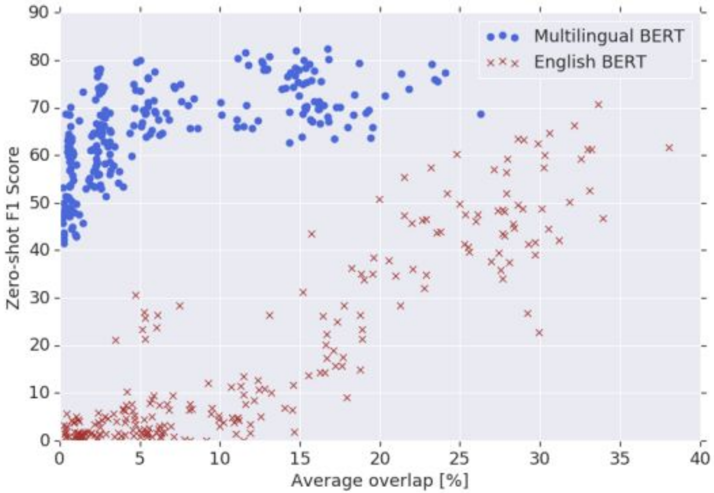
# Step 4: Adapting to downstream tasks under joint pretraining

**Basic Idea:** Fine-tune our multilingual language model on a task in language A and evaluate directly on the same task in language B

With a multilingual base model, we can get good performance on a task in language A by fine-tuning on that task in language B.

## mBERT Result (2019)[15]

- Fine-tune on a task in English, evaluate zero-shot in 40+ other languages → non-trivial performance




*"Zero-shot Named Entity Recognition F1 score versus entity word piece overlap among 16 languages. While performance using English BERT depends directly on word piece overlap, M-BERT's performance is largely independent of overlap, indicating that it learns multilingual representations deeper than simple vocabulary memorization."*

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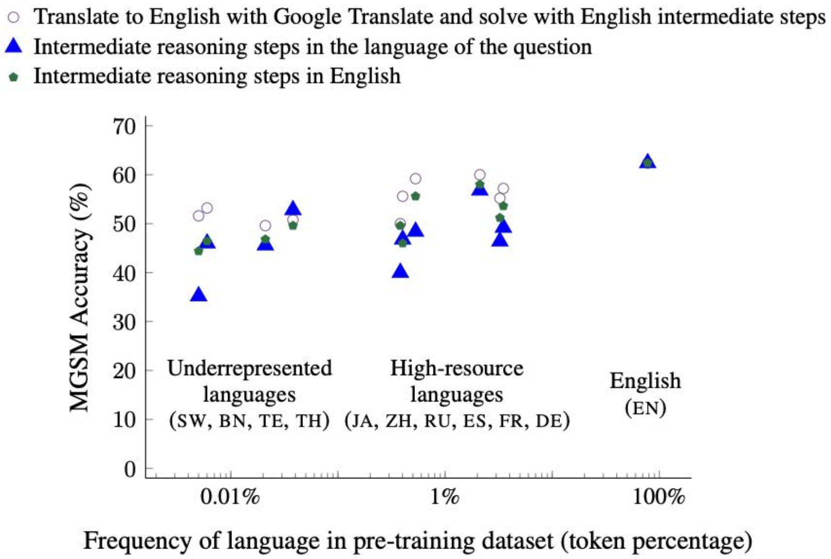
## Modern LLM analogue[16]

- Instruction-tune multilingual LLM in (mostly) English, get reasonable zero/few-shot performance across many languages at inference. Even on reasoning tasks!



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**Where It Falls Apart:**

- Transfer success correlates with script overlap, morphological similarity, and shared subword tokens
- Often optimal performance isn't achieved; can sometimes get away with only a bit of multilingual data[17] but often need to use adaptation methods from Route 1
- Transfer assumes the task is culturally invariant. Many task definitions ("positive sentiment", "toxic", "polite", "appropriate") differ across cultures; the fine-tuned model inherits the definitions and biases from the source language's culture

# Revisiting "The Curse of Multilinguality"

"The Curse" is a bit more nuanced than it seems. On downstream tasks, uniformly multilingual pretraining mixtures underperform mixtures dominated by one language.

## Why?

- Hypothesis: A dominant language provides a scaffold of world knowledge and reasoning skills that transfers across languages
- Some (debated) work suggests intermediate computations in English-heavy models look as if they pass through English[18]

## Practical takeaway

- Balanced data isn't necessarily the right choice. Some asymmetry seems to be what makes joint training work at all

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| Idea | Strong monolingual models: either (a) Train language-specific models from scratch or (b) Adapt a high-resource base model | One multilingual model: trained jointly on all languages from the start |
| What it exploits | Linguistic universals; reusable world knowledge in the base model; transferability of lower-layer representations | Shared parameters and vocabulary force the model to find common structure |
| Strengths | Each language gets a dedicated, optimized model; high baseline when target-language data is sufficient | One artifact handles all languages; cross-lingual tasks work natively; no per-language pipeline |
| Limitations | Substantial target-language data still needed; no sharing of task knowledge across languages; catastrophic forgetting on adaptation | Capacity dilution; the curse of multilinguality |
| Examples | CamemBERT (FR), FinBERT, Japanese Stable LM, Swallow, Sabia (PT), language-specific Llama variants | mBERT, XLM-R, BLOOM, Aya, Llama-3, EuroLLM, Apertus |

# Evaluation: what we (attempt to) measure

## Tasks with Multilingual Evaluations

- Language understanding & inference: XNLI, XCOPA, BELEBELE
- Factual knowledge: MMLU translations, Global-MMLU, INCLUDE (Romanou et al. 2024, EPFL)
- Reasoning: MGSM, mCoT-Math
- Generation: FLORES (translation), XLSum (summarization), Aya / MT-Bench-multi (instruction following)

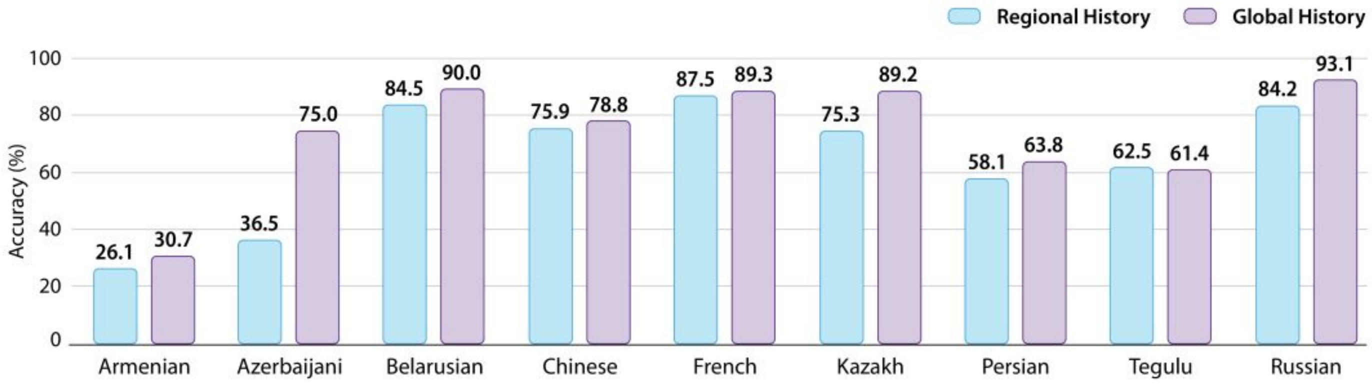
## Where it breaks

- Most benchmarks are translated from English; models can score well by reverse-translating internally. Benchmark scores correlate strongly with translation quality (Pearson $r \approx 0.87$–$0.91$ for Phi-4 across MT quality metrics)[19], suggesting they may largely be measuring translation ability
- Many evals don't check the response language; English-biased models get partial credit for English answers to non-English prompts[20]
- Translated benchmarks themselves contain translation errors and translationese

# Evaluation: what we don't measure

Cultural reasoning and regional knowledge

- Some reasoning depends on culturally specific knowledge: legal systems, holidays, naming conventions, religious practice
- Most multilingual evaluation tests whether the model can do English-content tasks in language X, not what users of language X actually need[21]




GPT-4o performance (In-language Prompt) on regional history exams (cultural) and global history exams from that region (region-implicit)

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## Code-switching

- Real users in many regions mix languages within a single sentence (Hindi-English, Spanish-English, Swahili-English)
- Almost no widely used benchmarks measure robustness to this setting; in practice, models greatly underperform

## Dialect

- Coverage within "supported" languages is uneven: MSA vs regional Arabics, Standard vs Swiss German
- Dialect speakers often get worse performance even when the standard form is supported

## The long tail of languages

- For most of the ~7,000 living languages, no usable benchmark exists at all. Performance for those languages is essentially unknown

# Open problems and recap

## What we covered

- Different routes to multilingual NLP
  - Adapting monolingual base models
  - The multilingual model training recipe: data → tokenization → joint pretraining → adaptation
- Cross-lingual transfer
- Curse of multilinguality: capacity vs. coverage trade-off
- Evaluation and its gaps

## What's still open in multilingual NLP

- Hundreds of millions of speakers without serious LLM coverage; Apertus and similar are early steps. Coverage is not the same as performance
- Equitable tokenization across scripts and morphology
- Low-resource adaptation without catastrophic forgetting
- Evaluating real-world utility, not just leaderboard scores
- Regional vs. translated benchmarks (INCLUDE-style)
- Dialects, registers, and code-switching robustness

*The gap between "N-language support" and equitable utility is one of the field's biggest unsolved problems.*

# Topics our lab is working on

- Multilingual LLM architectures
- Multilingual evaluation
- Language identification
- Multilingual tokenization

*Join us!*

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